

RESEARCH PROPOSAL TEMPLATE

Research title <i>(English)</i>	Change-Point-Guided Validation of TTF Stage Boundaries for Three-Stage Bearing Health Classification
Sub-committee	Information Technology (IT)
Group name	SE1940-G05
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Abstract

This study proposes a framework to validate the widely used Time-to-Failure (TTF) boundaries (e.g., 0.60/0.90) for categorizing bearing run-to-failure data into Healthy, Degrading, and Fault stages. By integrating multi-modal health indicators from vibration and temperature telemetry, utilizing change-point detection (CPD) algorithms, and performing downstream classification robustness analysis, this research aims to provide a data-driven justification for stage labeling.

Introduction

1.1. Literature review

In Prognostics and Health Management (PHM), assigning accurate degradation stages is critical for training robust classification models. While existing literature focuses heavily on predicting Remaining Useful Life (RUL), Change-Point Detection (CPD) is rarely used to validate heuristic TTF thresholds for standard 3-class classification tasks

1.2. The necessity of the research

Dividing bearing lifecycles using fixed heuristic boundaries frequently introduces label noise. It is necessary to develop a validation framework assessing threshold sensitivity to confirm if standard 60/90 TTF thresholds are optimal or if data-driven alternatives are required.

1.3. Feasibility of Research

The proposed study implements a software-based pipeline using public run-to-failure datasets. It requires no hardware funding and relies on open-source Python libraries (such as scikit-learn and ruptures) for feature extraction and algorithm implementation.

2. Research Objectives

- 1. To construct multi-modal Health Indicators (HI) by fusing vibration and temperature features.*
 - 2. To apply Change-Point Detection (CPD) methods to identify natural transition regions correlating with TTF thresholds.*
 - 3. To evaluate label sensitivity and classification robustness across various threshold settings.*
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3. Research Scope

Inclusions: *Construction of HI via PCA, binary segmentation CPD, separability analysis, and classification validation using standard machine learning models.*

Exclusions: *Complex deep learning-based RUL predictions and physical hardware deployment are omitted.*

4. Approach and Method

1. **Feature Extraction:** Extracting vibration features (RMS, Kurtosis, Spectral Energy) and temperature features (mean, delta, slope).
2. **Health Indicator Construction:** Employing Principal Component Analysis (PCA) to fuse signals into a distinct degradation trajectory.
3. **Change-Point Detection:** Executing CPD algorithms (Binary Segmentation/PELT) to isolate phase transitions computationally.
4. **Robustness Validation:** Training classifiers (e.g., SVM) across multiple threshold combinations to assess classification stability (focusing on Macro-F1 scores).

5. Research plan

No.	Date	Task	Output
1	Week 1	Prepare metadata, process signal data, and extract features.	Feature extraction scripts and CSV matrices
2	Week 2	Construct Vibration-Temperature Health Indicators (HI).	HI trajectory visualizations
3	Week 3	Implement Change-Point Detection algorithms.	CPD isolation boundaries
4	Week 4	Train classifiers and conduct sensitivity analysis.	Robustness metrics and Boundary Scores

5	Week 5	Compile charts, format data, and write final report.	Final paper draft
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Expected Result:

- ***Theoretical implications:*** Proves that boundary verification yields highly reliable health state labels.
- ***Methodological implications:*** Delivers a clear pipeline integrating multi-modal HI generation, CPD, and sensitivity analysis.
- ***Practical implications:*** Supplies robust justification for the usage of conventional TTF limits in industrial edge diagnostics.

References:

- [1] T. A. Author, "Stage-Based Remaining Useful Life Prediction for Bearings," *Machines*, 2025.
- [2] B. B. Author, "A Change Point Detection Integrated Remaining Useful Life Estimation Model," *arXiv*, 2024.

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We are authors of the research paper titled “**Reliable IoT Sensor Data Transmission using Hamming Code (7,4)**”, would like to ensure that all members comply with the rules relevant to Artificial Intelligence (AI) regulations and ethical guidelines in research practices, which are described in details as follows:

1. AI is only used as assistant in correcting grammar, structure and phrasing in academic writing;
2. AI usage should NOT be engaged with the creation of content or drafting the research paper, production of fake data or literature synthesis;
3. AI usage should be checked and confirmed by all authors. Each author shall bear full responsibility in the accuracy, transparency and originality of their sections, including citations and reference;

4. All authors should disclose the use of generative AI and AI-assisted technologies to other co-authors, helping them understand where and how AI was applied in research preparation.

Note:

- *Research proposal must be presented in English/Japanese, A4 paper, single column, Times New Roman 11, and a maximum length of 10 pages (excluding references and appendices)*
- *For citations and references, please refer to:*

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